QPCODE : 24900064

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMEBER 2024

FIRST SEMESTER - MG1CCRBCA100 DIGITAL FUNDAMENTALS

MAX MARKS 70

Part A – Answer all questions – each question carries 2 marks

1. Overflow in binary addition occurs when the sum of two binary numbers exceeds the range that can be represented with the given number of bits. [U][1]

2. Truth table of half adder -2 marks [U][3]

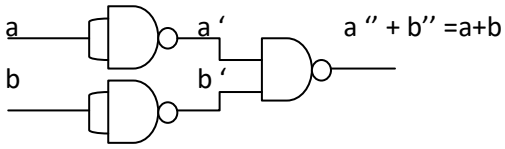
3. Separate the decimal number into individual digits .Write the BCD equivalent of each digit and combine them . [U][1]

4. Truth table of RS flip-flop 1 mark. Explanation 1 mark [An][4]

5. BCD addition - 1001 1000 - 2 marks [A] [1]

6. A NAND gate can act as an OR gate by using De Morgan's Theorem and by combining multiple NAND gates in a specific configuration. [U][2]

   (a ' . b ')'=a '' + b '' = a+b



7. f = AB+ B'C=AB(C+C')+B'C(A+A')=ABC+ABC'+AB'C+A'B'C [A] [2]

8. Sequential circuit definition or block diagram 1 mark, Explanation 1 mark [U][4]

9. Any two use of flip flops. Data storage, counters, frequency division, shift registers -2 marks [U][4]

10. Logic circuit of 3-to-8 decoder 2 marks [U][3]

Part B – Answer any 5- each question carries 6 marks

11. – binary subtraction of 10101101-01011011 Ans.01010010 : 3marks, discussion and steps 3marks [A] [1]

12. Simplification using Boolean laws 3 marks each [A] [2]

    (i) F=AB+A(B+C)+B(B+C)=AB+AB+AC+BB+BC=AB+AC+B+BC=AB+AC+B(1+C)

    =AB+AC+B=AC+A(B+1)=AC+A=A(C+1)=A

    (ii) F=A'B+BC'+BC+AB'C'=A'B+B(C+C')+AB'C'=A'B+B+AB'C'=B(A'+1)+AB'C'=B+AB'C'

13. Design 8x1 multiplexer using 4x1 and 2x1 multiplexer -6marks [A] [3]

    Components:

    a) Two 4x1 multiplexers to handle 8 input lines.
    b) One 2x1 multiplexer to select between the outputs of the two 4x1 multiplexers.

14. Truth table of RS flip-flop 2 marks, circuit diagram 2 marks explanation 2 marks [U][4]

15. 256(Decimal)=001001010110(BCD) 3marks explanation with steps 3 marks [A] [1]

16. Application of De Morgan's theorem 2 marks each [A] [2]

    (i) [(A+B+C)D]'=(A+B+C)'+D'= (A'·B'·C')+D'= A'·B'·C'+D'
    (ii) (ABC+DEF)'=(ABC)'·(DEF)'= (A'+B'+C')·(D'+E'+F')
    (iii) [AB'+C'D+EF]'= (AB')'·(C'D)'·(EF)' =(A'+B)·(C+D')·(E'+F')

17. Circuit diagram, graphic symbol and truth table for D-fli flop – 2 marks each [U][4]

Part C– Answer any 2- each question carries 10 marks

18. Basic laws of Boolean Algebra -5 marks. Simplification Steps 5marks [A] [2]

    F=AB+A(B+C)+B(B+C)= AB+AB+AC+BB+BC= AB+AC+B+BC=AB+AC+B(1+C)

    =AB+AC+B=B(A+1)+AC=B+AC

19. RS flip-flop - Working with truth table (3marks) circuit diagram(2marks) [U][4]

    JK flip-flops -Working with truth table (3marks) circuit diagram(2marks)

20. Explain the working of an encoder with a suitable diagram 6 marks, Applications and advantages 4 marks [U][3]